Topic 1: What is Interaction Design (part 2)

SCSM2113 Human-Computer Interaction Fundamentals

Faculty of Computing
Universiti Teknologi Malaysia

1

01 GOOD AND BAD DESIGN

02 WHAT IS INTERACTION DESIGN

03 UNDERSTANDING PEOPLE

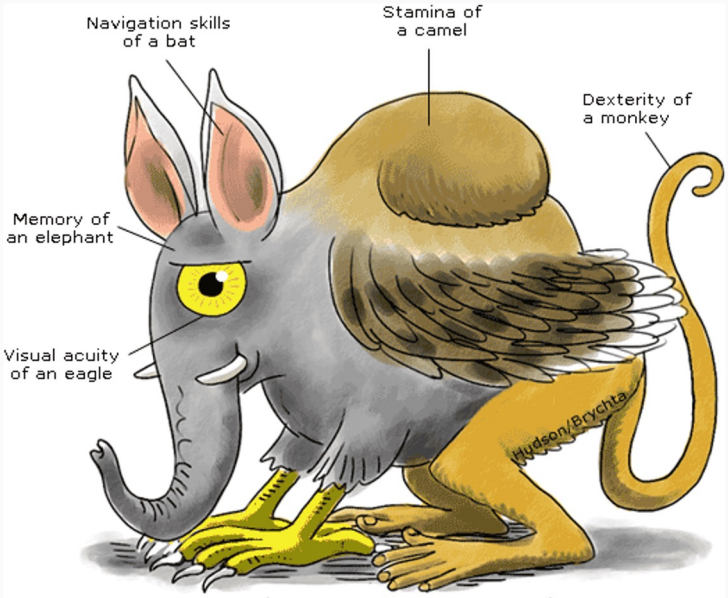
04 USABILITY & USER EXPERIENCE GOALS

05 DESIGN PRINCIPLES

2

GOOD & BAD DESIGN

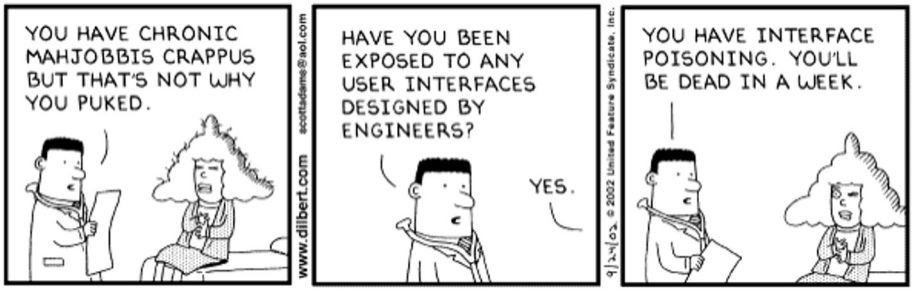
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"The Perfect User"

4





5

Bad Design: Lift Control

- Lift controls and labels on the bottom row all look the same, so it is easy to push a label by mistake instead of a control button



Source: www.baddesigns.com

- People do not make same mistake for the labels and buttons on the top row. Why not?

6

Bad Design: Why is this vending machine so bad?



Source: www.baddesigns.com

- Need to **push button first** to activate reader
- Normally **insert bill** first before making selection
- Contravenes well known convention

Good Design

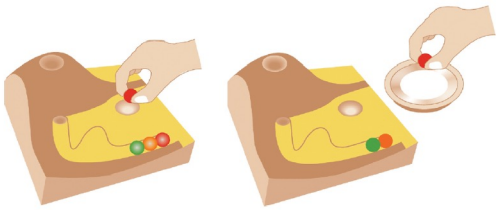
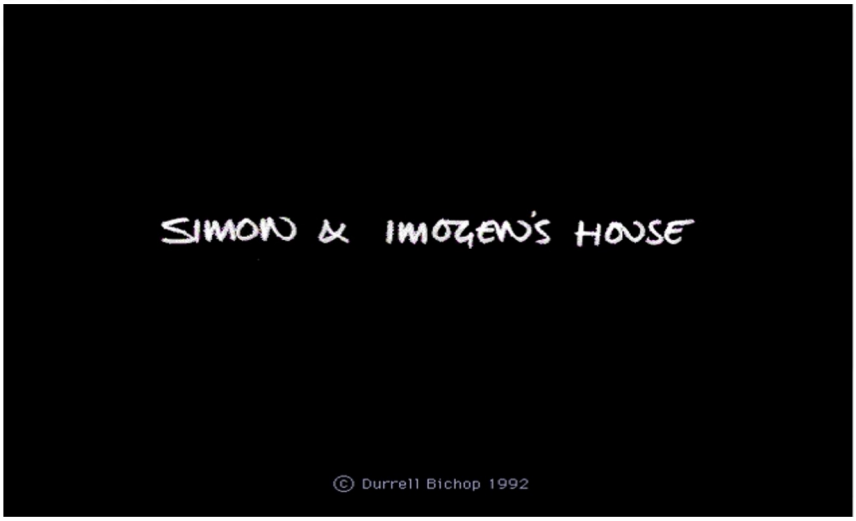


Figure 1.1 The marble answering machine
Source: Adapted from Gillian Crampton Smith: "The Hand that Rocks the Cradle" *ID Magazine*, May/June 1995, pp. 60–65.

- Marble answering machine (Bishop, 1995)
- Based on how everyday objects behave
- Easy, intuitive and a pleasure to use
- Only requires one-step actions to perform core tasks



How does the "marble" answering machine differ from the voice mail system?

9

9

Why is it a Good Design?

- Use familiar physical object "marble pinball" that visually represent how many incoming message been left.
- Play recorder message
  - dropping the marble into a slot in the machine
- Dial the caller who left the message
  - Dropping the same marble pinball into another slot on phone
- It is aesthetically pleasing and enjoyable to use
- It only required one-step actions to perform core tasks
- It is a simple but elegant design
- It offers less functionality and allow anyone to listen to any the message

10

10

Good and Bad Design

- Why is the TiVo remote much better designed than standard remote controls?
  - Peanut shaped to fit in hand
  - Logical layout and color-coded, distinctive buttons
  - Easy-to-locate buttons



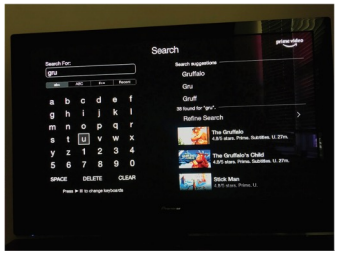


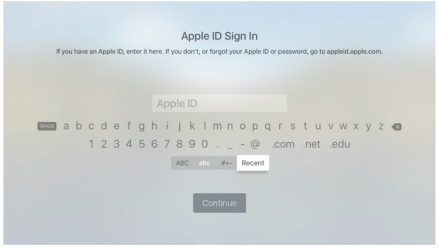
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11

Dilemma

- Which is the best way to interact with a smart TV? Why?
- Pecking using a grid keyboard via a remote control
- Swiping across two alphanumeric rows using a touchpad on a remote control
- Voice control using remote or smart speaker





12

12

Switching to Digital

- Many activities that used to be done with a physical artifact have gone digital
  - e.g. ticket machines, parking meters
- Benefit
  - Makes many tasks easier, quicker and more convenient
  - Our details can also be stored ready for next time
  - Don't have to wait in line before when buying a physical ticket
  - Easy to swipe QR code to gain entrance through a ticket barrier
- Disadvantages
  - Not everyone has a modern smartphone
  - Some people still prefer to use older phones which the apps won't work on
  - Some people prefer not to divulge personal information online
  - Some people prefer to talk to real people when making a purchase

What to Design

- Need to take into account:
  - Who the users are
  - What activities are being carried out
  - Where the interaction is taking place
- Need to optimise the interactions users have with a product:
  - So that they match the users' activities and needs

Understanding User's Needs

- Take into account what people are good and bad at
- Consider what might help people in the way they currently do things
- Think through what might provide quality user experiences
- Listen to what people want and get them involved
- Use tried and tested user-centered methods

15

WHAT IS INTERACTION DESIGN

16

What is Interaction Design?

- "Designing interactive products to support the way people communicate and interact in their everyday and working lives."

Preece, Sharp and Rogers (2015)

- "The design of spaces for human communication and interaction."

Winograd (1997)

17

Goals of Interaction Design?

- Develop usable products
  - Usability means easy to learn, effective to use and provide an enjoyable experience
- Involve users in the design process

18

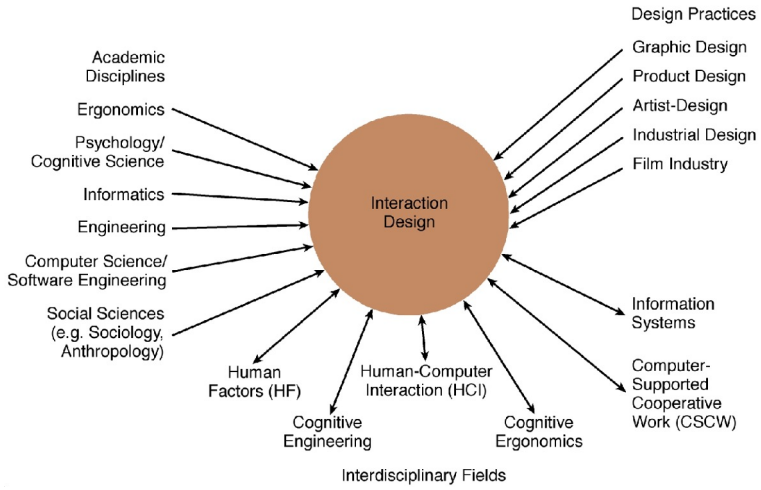
Which kind of Design?

- Number of other terms used emphasising what is being designed, e.g.
  - user interface design, software design, user-centered design, product design, web design, experience design (UX)

- Interaction design is the umbrella term covering all of these aspects
  - fundamental to all disciplines, fields, and approaches concerned with researching and designing computer-based systems for people

19

HCI and Interaction Design



20

Relationship between ID, HCI & Other Fields

Academic disciplines contributing to ID:

- Psychology
- Social Sciences
- Computing Sciences
- Engineering
- Ergonomics
- Informatics

Design practices contributing to ID:

- Graphic design
- Product design
- Artist-design
- Industrial design
- Film industry

Relationship between ID, HCI & Other Fields (cont.)

Interdisciplinary fields that 'do' interaction design:

- HCI
- Ubiquitous Computing
- Human Factors
- Cognitive Engineering
- Cognitive Ergonomics
- Computer Supported Co-operative Work
- Information Systems

Working in Multidisciplinary Teams

- Many people from different backgrounds involved
- Different perspectives and ways of seeing and talking about things

- Benefits
  - more ideas and designs generated
- Disadvantages
  - difficult to communicate and progress forward the designs being create

23

Interaction Design in Business

- Increasing number of ID consultancies, examples of well-known ones include:
  - Nielsen Norman Group: "help companies enter the age of the consumer, designing human-centered products and services"
  - Cooper: "From research and product to goal-related design"
  - Swim: "provides a wide range of design services, in each case targeted to address the product development needs at hand"
  - IDEO: "creates products, services and environments for companies pioneering new ways to provide value to their customers"
  - *Stampede: "Transforming online experiences for business and organisations with incredible, innovative solution since 2006"*

24

UNDERSTANDING
PEOPLE

25

People-centered Design

- Involves understanding how people feel about a product and their pleasure and satisfaction when using it, looking at it, holding it, and opening or closing it.
- Their overall impression of how good it is to use
- The quality of the experience
- *"It is not enough that we build products that function, that are understandable and usable, we also need to build joy and excitement, pleasure and fun, and yes, beauty to people's lives."*
  - Don Norman (2004)

26

The User Experience (UX)

- How a product behaves and is used by people in the real world
  - the way people feel about it and their pleasure and satisfaction when using it, looking at it, holding it, and opening or closing it
  - "every product that is used by someone has a user experience: newspapers, ketchup bottles, reclining armchairs, cardigan sweaters." (Garrett, 2010)
  - "all aspects of the end-user's interaction with the company, its services, and its products. (Nielsen and Norman, 2014)

- Cannot design a user experience, only design ***for*** a user experience

27

Defining UX

- How users perceive a product, such as whether a smartwatch is seen as sleek or chunky, and their emotional reaction to it, such as whether people have a positive experience when using it.

  (Hornbæk and Hertzum, 2017)

- Hassenzahl's (2010) model of the user experience
  - Pragmatic: how simple, practical, and obvious it is for the user to achieve their goals
  - Hedonic: how evocative and stimulating the interaction is to users

28

Why was the iPod UX such a success?



Figure 1.6 The iPod Nano Touch
Source: ©Press Association, reproduced with permission.

- Quality user experience from the start
- Simple, elegant, distinct brand, pleasurable, must have fashion item, catchy names, cool...

29

29

and today..







Spotify

RUNCLOUD

NETFLIX

30

30

Core Characteristics of Interaction Design

- Users should be involved through the development of the project
- Specific usability and user experience goals need to be identified, clearly documented and agreed at the beginning of the project
- Iteration is needed through the core activities

Why?

- Help designers:
  - understand how to design interactive products that fit with what people want, need and may desire
  - appreciate that one size does not fit all
    - e.g., teenagers are very different to grown-ups
  - identify any incorrect assumptions they may have about particular user groups
    - e.g., not all old people want or need big fonts
  - be aware of both people's sensitivities and their capabilities

Accessibility and Inclusiveness

- ***Accessibility***: the extent to which an interactive product is accessible by as many people as possible
  - Focus is on people with disabilities; for instance, those using Android OS or apple voiceover

- ***Inclusiveness***: making products and services that accommodate the widest possible number of people
  - For example, smartphones designed for all and made available to everyone regardless of their disability, education, age, or income

Disabilities

- Whether someone is disabled changes over time with age, or recovery from an accident

- The severity and impact of an impairment can vary over the course of a day or in different environmental conditions

- Disabilities can result because technologies are designed to necessitate a certain type of interaction that is impossible for someone with an impairment

Understanding Disabilities

- Disabilities can be classified as:
  - Sensory impairment (such as loss of vision or hearing)
  - Physical impairment (having loss of functions to one or more parts of the body after a stroke or spinal cord injury)
  - Cognitive (including learning impairment or loss of memory/cognitive function due to old age)
- Each type can be further defined in terms of capability:
  - For example, someone might have only peripheral vision, be color blind, or have no light perception
- Impairment can be categorised:
  - Permanent (for instance, long-term wheelchair user)
  - Temporary (that is, after an accident or illness)
  - Situational (for example, a noisy environment means that a person can't hear)

35

Being *Cool* about Disabilities

- Prosthetics can be designed to move beyond being functional (and often ugly) to being desirable and fashionable
- People now refer to "wearing their wheels," rather than "using a wheelchair"



Fashionable leg cover designed by Alleles Design Studio

36

USABILITY & USER EXPERIENCE GOALS

37

Usability Goals

- Effective to use
- Efficient to use
- Safe to use
- Have good utility
- Easy to learn
- Easy to remember how to use

38

User Experience Goals

Desirable aspects

satisfying	helpful	fun
enjoyable	motivating	provocative
engaging	challenging	surprising
pleasurable	enhancing sociability	rewarding
exciting	supporting creativity	emotionally fulfilling
entertaining	cognitively stimulating	

Undesirable aspects

boring	unpleasant
frustrating	patronising
making one feel guilty	making one feel stupid
annoying	cutesy
childish	gimmicky

39

39

Usability and UX Goals

- Selecting terms to convey a person's feelings, emotions, etc., can help designers understand the multifaceted nature of the user experience
- How do usability goals differ from user experience goals?
- Are there trade-offs between the two kinds of goals?
  - e.g. can a product be both fun and safe?
- How easy is it to measure usability versus user experience goals?

40

40

DESIGN PRINCIPLES

41

Design Principles

- Generalisable abstractions for thinking about different aspects of design
- The do's and don'ts of interaction design
- What to provide and what not to provide at the interface
- Derived from a mix of theory-based knowledge, experience and common-sense

42

Visibility



www.baddesigns.com

- This is a control panel for a lift
- How does it work?
- Push a button for the floor you want?
- Nothing happens. Push any other button? Still nothing. What do you need to do?

- It is not visible as to what to do!

43

Visibility (cont.)



www.baddesigns.com

...with this lift, you need to insert your room card in the slot by the buttons to get the lift to work!

How would you make this action more visible?

- make the card reader more obvious
- provide an auditory message, that says what to do (which language?)
- provide a big label next to the card reader that flashes when someone enters
- make relevant parts visible
- make what has to be done obvious

44

Visibility (cont.)



What do I do if I am wearing black?

Invisible automatic controls can make it more difficult to use

Feedback

- Sending information back to the user about what has been done
- Includes sound, highlighting, animation and combinations of these
  - e.g. when screen button clicked on provides sound or red highlight feedback:

Previous → "ccclichhk"

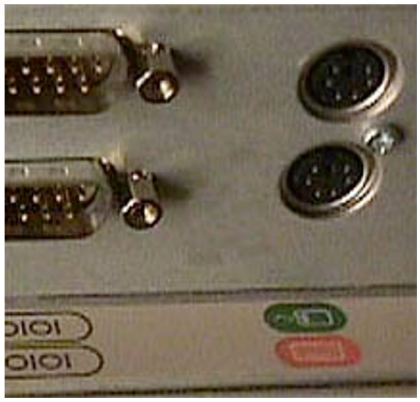
Previous → Previous

Constraints

- Restricting the possible actions that can be performed
- Helps prevent user from selecting incorrect options
- Physical objects can be designed to constrain things
  - e.g. only one way you can insert a key into a lock

47

Constraints (cont.)



www.baddesigns.com

Logical or ambiguous design?

- Where do you plug the mouse?
- Where do you plug the keyboard?
- top or bottom connector?
- Do the colour-coded icons help?

48

Constraints (cont.)



www.baddesigns.com

How to design them more logically?

(i) A provides direct adjacent mapping between icon and connector

(ii) B provides colour coding to associate the connectors with the labels



www.baddesigns.com

Consistency

- Design interfaces to have similar operations and use similar elements for similar tasks
- For example:
  - always use ctrl key plus first initial of the command for an operation – ctrl+C, ctrl+S, ctrl+O
- Main benefit is consistent interfaces are easier to learn and use

Consistency (cont.)

When consistency breaks down

- What happens if there is more than one command starting with the same letter?
  - e.g. save, spelling, select, style
- You have to find other initials or combinations of keys, thereby breaking the consistency rule
  - e.g. ctrl+S, ctrl+Sp, ctrl+shift+L
- Increases learning burden on user, making them more prone to errors

Consistency (cont.)

Internal and external consistency

- Internal consistency refers to designing operations to behave the same within an application
  - Difficult to achieve with complex interfaces
- External consistency refers to designing operations, interfaces, etc., to be the same across applications and devices
  - Very rarely the case, based on different designer's preference

Consistency (cont.)

Keypad numbers layout

- A case of external inconsistency

(a) phones, remote controls

1	2	3
4	5	6
7	8	9
	0	

(b) calculators, computer keypads

7	8	9
4	5	6
1	2	3
0		

Affordance: to give clue

- Refers to an attribute of an object that allows people to know how to use it
  - e.g. a mouse button invites pushing, a door handle affords pulling
- Norman (1988) used the term to discuss the design of everyday objects
- Has since popularised in interaction design to discuss how to design interface objects
  - e.g. scrollbars to afford moving up and down, icons to afford clicking on

Affordance (cont.)

What does 'affordance' have to offer interaction design?

- Interfaces are virtual and do not have affordances like physical objects
- Norman argues it does not make sense to talk about interfaces in terms of 'real' affordances
- Instead interfaces are better conceptualised as 'perceived' affordances
  - Learned conventions of arbitrary mappings between action and effect at the interface
  - Some mappings are better than others

55

UTM

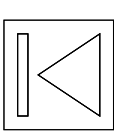
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Activity

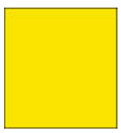


Virtual affordances

- How do the following screen objects afford?
- What if you were a novice user?
- Would you know what to do with them?









56

UTM

56

Shneiderman's Eight Golden Rules

The principles	Questions to consider
1. Strive for consistency	Is the style of this element maintained across your site/app? Is this content placed in the correct location according to the site hierarchy? Does this follow the conventions for your chosen platform? How can you make your designs more consistent?
2. Enable frequent users to use shortcuts	Are there shortcuts available for your more experienced users? Who is this product designed for? Will there be a need to consider experienced users? How can you make it easier and quicker for experienced users?
3. Offer informative feedback	Does the user know where they are at in the process? Does the user know what they have done after performing this action? How are you communicating this feedback to your user?
4. Design dialogue to yield closure	Does the user have to do any guessing here? Is it clear and obvious enough for your intended audience? Are there any next steps for the user? How are you communicating the system status with the user?
5. Offer simple error handling	Have you done everything imaginable to prevent this error from happening on your end? Is this error avoidable in the first place? If the user does make an error, how easy is it for them to fix it?
6. Permit easy reversal of actions	How many steps does the user have to take to reverse their actions? Will the user quickly realize they need to reverse the action in the first place? How can you make your users detect the possibility of reversal?
7. Support internal locus of control	Will the user feel in control at this specific touch point in your app? Will they be surprised in an unpleasant manner? Does the site feel easily navigable? Does the user feel safe and in control? How can you make the user feel more safe and in control??
8. Reduce short-term memory load	Are there enough visual cues here for the user to find the functionality or item? Do they have to remember things to understand what's going on? How can you help the user recall?

57

Key Points

- Interaction design is concerned with designing interactive products to support how people communicate and interact in their everyday and working lives
- It is concerned with how to create quality user experiences for services, devices, and interactive products
- It is multidisciplinary, involving many inputs from wide-reaching disciplines and fields
- Optimising the interaction between users and interactive products requires consideration of a number of interdependent factors, including context of use, types of activity, UX goals, accessibility, cultural differences, and user groups.
- Design principles, such as feedback and simplicity, are useful heuristics for informing, analysing, and evaluating aspects of an interactive product.

58



59